

EC601 — Control System

Semester VI • 3L:0T:0P • 3 Credits

Module 7 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

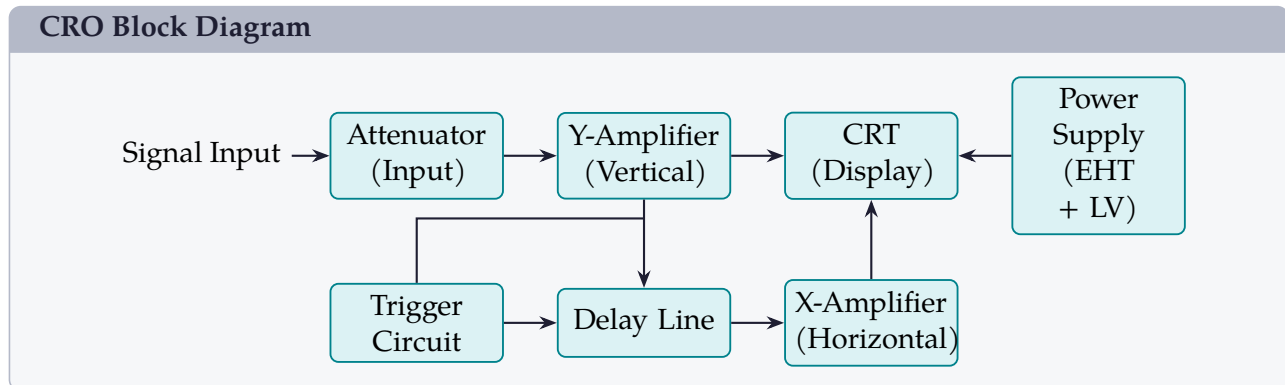
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1. Cathode Ray Oscilloscope (CRO)

CRO (Cathode Ray Oscilloscope): An electronic test instrument that displays electrical signals as waveforms on a phosphor screen. It plots voltage (Y-axis) vs time (X-axis) in real time, allowing measurement of amplitude, frequency, phase, rise time, and waveform shape.

1.1 Block Diagram of CRO



1.2 Main Blocks and Functions

Block	Function
Attenuator	Reduces input signal to a safe level for the amplifier. Calibrated in V/div.
Y-Amplifier	Amplifies the vertical (signal) input to drive the vertical deflection plates of CRT.
Delay Line	Delays the signal slightly so the sweep starts before the signal arrives — ensures complete waveform display.
Trigger Circuit	Synchronises the sweep with the input signal to produce a stable display.
Time Base (Sweep Gen.)	Generates a sawtooth waveform for horizontal deflection. Calibrated in time/div.
X-Amplifier	Amplifies the time base signal to drive the horizontal deflection plates.
CRT	Cathode Ray Tube — electron gun, deflection plates, phosphor screen. Displays the waveform.
Power Supply	Provides EHT (~1–10 kV) for CRT and low voltages for amplifiers.

1.3 Measurements with CRO

Measurement	Method
Voltage (peak)	Count vertical divisions $\times$ V/div setting. $V_{pk} = n_{div} \times V/div$
Time period T	Count horizontal divisions for one cycle $\times$ time/div. $T = n_{div} \times \text{time/div}$
Frequency	$f = 1/T$
Phase difference	Use dual-trace mode; measure time shift Δt . $\phi = (\Delta t/T) \times 360^\circ$
Rise time	Time for signal to go from 10% to 90% of final value.
DC voltage	Set coupling to DC; measure vertical displacement from zero.

1.4 Lissajous Figures (Phase & Frequency Measurement)

Lissajous figures are patterns displayed on a CRO screen when two sinusoidal signals of different or same frequencies are applied simultaneously to the horizontal (X) and vertical (Y) deflection plates. They are used to measure **phase difference** and **frequency ratio**.

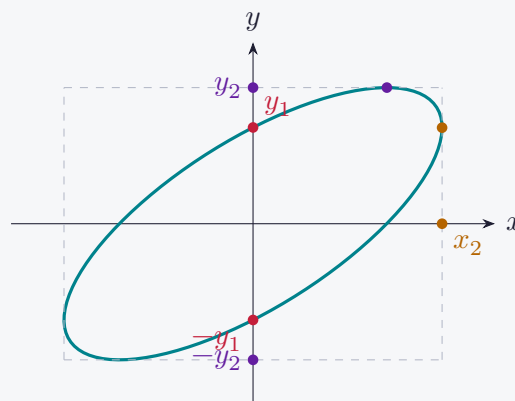
Phase Difference Measurement

When two sine waves of equal frequency ($f_x = f_y$) but different phase are applied, the pattern is an **ellipse**. The phase difference θ is calculated from the intercepts:

$$\sin \theta = \frac{y_1}{y_2} \Rightarrow \theta = \sin^{-1}\left(\frac{y_1}{y_2}\right) \quad \text{or} \quad 180^\circ - \sin^{-1}\left(\frac{y_1}{y_2}\right)$$

where y_1 is the y-intercept (height at $x = 0$) and y_2 is the maximum vertical deflection.

Lissajous Phase Ellipse



Frequency Measurement

If the frequencies are unequal, the frequency ratio is determined by drawing a horizontal and a vertical line through the pattern and counting the number of tangency points:

$$\frac{f_y}{f_x} = \frac{\text{Number of Horizontal Tangencies } (N_h)}{\text{Number of Vertical Tangencies } (N_v)}$$

1.5 Special Purpose Oscilloscopes: Dual-Trace vs. Dual-Beam & DSO

Modern electronic testing requires displaying multiple signals simultaneously and storing waveforms for analysis.

1.5.1 Dual-Trace vs. Dual-Beam Oscilloscopes

Parameter	Dual-Trace CRO	Dual-Beam CRO
Electron Gun	Single electron gun.	Two separate electron guns (or split beam with two deflection systems).
Deflection System	One set of vertical deflection plates.	Two independent sets of vertical deflection plates.
Working Method	Uses electronic switching (Chop or Alternate mode) to display two traces.	Displays two signals simultaneously by writing with two separate beams.
Modes	Chop: switches channels at high speed (low f signals). Alternate: swaps channel after each sweep (high f signals).	Simpler circuitry; no switching required.
Cost & Complexity	Moderate cost; standard design.	Highly complex; expensive CRT assembly.
Trace Alignment	Perfect alignment; single electron path.	Beam alignment can drift over time.

1.5.2 Digital Storage Oscilloscope (DSO)

A **Digital Storage Oscilloscope (DSO)** digitises the analog input signal and stores it in digital memory. It then reads this data from memory to reconstruct and display the waveform on the screen.

DSO Digitisation Block Diagram



- **Sample and Hold (S&H):** Holds the analog voltage constant long enough for the ADC to convert it.
- **ADC (Analog-to-Digital Converter):** Converts the held analog sample into a digital word (binary code).
- **Digital Memory (RAM):** Stores the binary samples sequentially.
- **DAC (Digital-to-Analog Converter):** Converts the stored data back to an analog signal.
- **Display:** Reconstructs and plots the signal.

Key Advantages of DSO over Analog CRO:

1. **Infinite Storage Time:** Waveforms are stored in memory and can be held indefinitely.
2. **Pre-trigger Viewing:** DSOs can display signals that occurred *before* the trigger point using a circular buffer.
3. **Signal Processing:** Direct microprocessor analysis (FFT, peak-to-peak, RMS, frequency).
4. **Easy Export:** Stored data can be transferred to a computer.

2. Wave Analyzer and Spectrum Analyzer

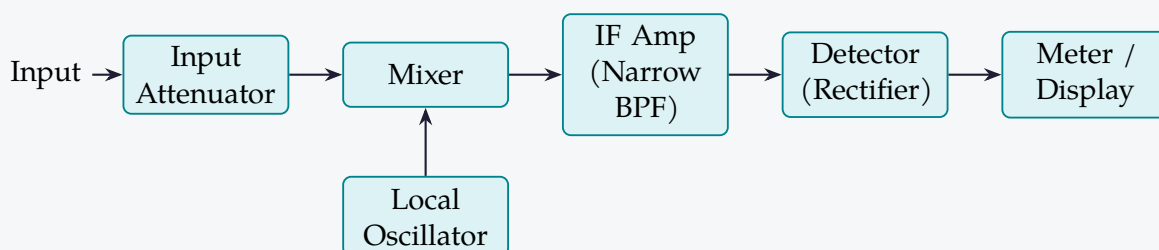
2.1 Wave Analyzer

Wave Analyzer: An instrument that measures the **amplitude** and **frequency** of individual harmonic components of a complex periodic waveform. It is essentially a *very narrow bandpass filter* that can be tuned to any frequency within its range.

Requirements:

- Very narrow, stable, and tunable bandpass filter
- High selectivity (ability to separate closely spaced harmonics)
- High sensitivity (detect small amplitude components)
- Calibrated frequency and amplitude readout

Wave Analyzer Block Diagram

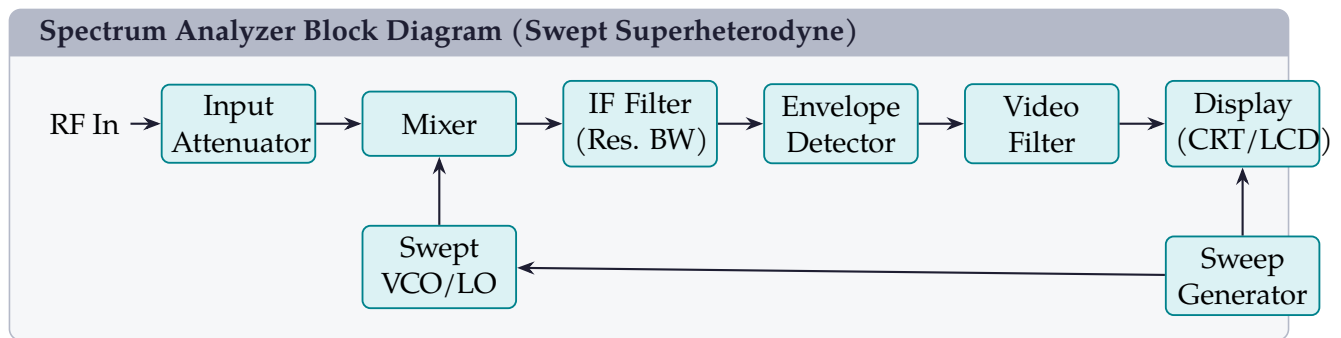


2.2 Spectrum Analyzer

Spectrum Analyzer: Displays the **frequency spectrum** of a signal — amplitude (or power) vs frequency. Unlike the wave analyzer (which measures one frequency at a time), the spectrum analyzer displays *all frequency components simultaneously* on a screen (frequency domain display).

Requirements:

- Wide frequency range with fine resolution
- High dynamic range (detect weak signals near strong ones)
- Fast sweep speed for real-time display
- Calibrated frequency (X-axis) and amplitude/power (Y-axis in dBm)



Feature	Wave Analyzer	Spectrum Analyzer
Display	One frequency at a time	All frequencies simultaneously
Domain	Frequency (selective)	Frequency (full spectrum)
Speed	Slow (manual tuning)	Fast (swept or FFT)
Use	Harmonic distortion analysis	Wideband signal analysis

3. LVDT (Linear Variable Differential Transformer)

LVDT: A Linear Variable Differential Transformer is an electromechanical transducer that converts linear displacement into a proportional AC voltage. It is the most widely used inductive displacement sensor due to its high accuracy, infinite resolution, and frictionless operation.

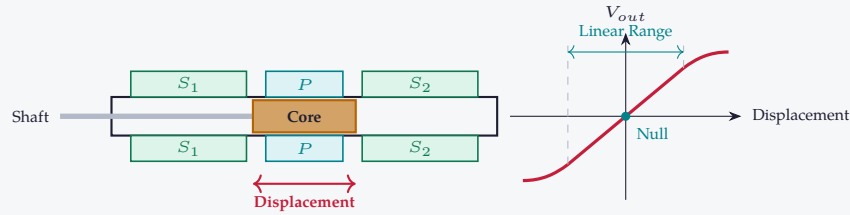
3.1 Construction and Working

Construction: One primary coil (centre) and two secondary coils (S_1 , S_2) wound symmetrically on a cylindrical former. A movable ferromagnetic **core** (plunger) slides inside without contact.

Working:

1. AC excitation applied to primary coil induces EMF in both secondary coils.
2. Secondary coils connected in **series opposition**: $V_{out} = V_{S1} - V_{S2}$.
3. **Null position** (core centred): $V_{S1} = V_{S2}$, so $V_{out} = 0$.
4. **Core displaced right**: $V_{S1} > V_{S2}$, V_{out} is positive and in phase with primary.
5. **Core displaced left**: $V_{S2} > V_{S1}$, V_{out} is negative (180° phase shift).

LVDT Construction and Output Characteristic



Advantages	Disadvantages
Infinite resolution (frictionless)	Sensitive to stray magnetic fields
High accuracy and repeatability	Requires AC excitation and demodulator
No mechanical wear (contactless)	Limited to linear displacement
Rugged; works in harsh environments	Bulkier than resistive sensors

4. Servomotors

Servomotor: A motor specifically designed for use in **closed-loop control systems** (servo systems). It must have a linear torque-speed characteristic, low inertia, fast response, and be controllable by an error signal. Used for precise position, speed, or torque control.

4.1 DC Servomotor

DC Servomotor: A DC motor with a separately excited field or permanent magnet field, controlled by varying the **armature voltage**. The torque is proportional to armature current; speed is proportional to back-EMF.

DC Servomotor Equations

Electrical: $V_a = E_b + I_a R_a + L_a \frac{dI_a}{dt}$, $E_b = K_b \omega$

Mechanical: $T = K_T I_a = J \frac{d\omega}{dt} + B \omega$

Transfer Function (armature control, $L_a \approx 0$):

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_T/R_a}{Js + B + K_T K_b/R_a}$$

where K_T = torque constant, K_b = back-EMF constant, J = inertia, B = damping.

Feature	DC Servomotor
Control method	Armature voltage control or field control
Torque-speed	Linear (ideal for servo use)
Response	Fast; low inertia rotor
Maintenance	Brushes require periodic replacement
Applications	Robotics, CNC machines, antenna positioning

4.2 AC Servomotor

AC Servomotor: A two-phase induction motor used in servo systems. One phase (reference phase) is supplied with a fixed AC voltage; the other phase (control phase) receives a variable voltage proportional to the error signal. Speed and direction are controlled by the magnitude and phase of the control voltage.

AC Servomotor Key Features

- **High rotor resistance:** Ensures stable, linear torque-speed characteristic (negative slope throughout).
- **Low inertia:** Drag-cup or thin-disc rotor for fast response.
- **Control:** Vary amplitude of control phase voltage; direction by phase reversal ($\pm 90^\circ$).
- **Transfer function:** $G(s) = K_m / (s(\tau_m s + 1))$ where K_m = motor gain, τ_m = mechanical time constant.

Feature	DC Servomotor	AC Servomotor
Supply	DC	2-phase AC
Control	Armature voltage	Control phase voltage
Maintenance	Brushes needed	Brushless (robust)
Power range	High power	Low to medium power
Noise	More (brushes)	Less
Applications	Heavy-duty servo	Instruments, light servo

5. Tachogenerator (Tacho Generator)

Tachogenerator: A transducer that converts **rotational speed** into a proportional electrical voltage. Used as a **velocity feedback sensor** in closed-loop speed control systems.

DC Tacho: Output DC voltage proportional to speed: $V_{out} = K_t \omega$

AC Tacho: Output AC voltage whose amplitude is proportional to speed; frequency

equals shaft rotation frequency.

Feature	DC Tachogenerator	AC Tachogenerator
Output	DC voltage $\propto$ speed	AC voltage amplitude $\propto$ speed
Direction sensing	Polarity reversal	Phase reversal
Ripple	Present (commutator)	None
Use	Speed feedback in DC drives	AC servo systems

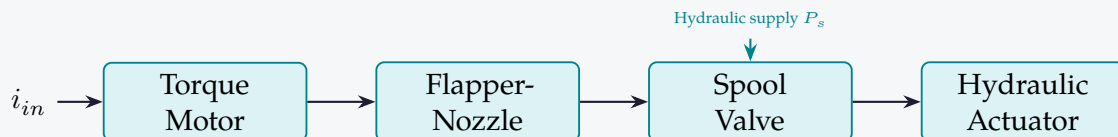
6. Hydraulic and Pneumatic Actuators

6.1 Electro-Hydraulic Valve

Electro-Hydraulic Valve (EHV): Converts an **electrical signal** (current or voltage) into a **hydraulic flow** or pressure. It controls the flow of hydraulic fluid to an actuator. Types: servo valve (proportional), solenoid valve (on-off).

Working: An electrical torque motor deflects a flapper between two nozzles, creating a differential pressure that positions a spool valve, which controls hydraulic flow to the actuator.

Electro-Hydraulic Servo Valve (Flapper-Nozzle Type)



6.2 Hydraulic Servomotor

Hydraulic Servomotor: A hydraulic actuator (linear cylinder or rotary motor) that converts **hydraulic power** (pressure $\times$ flow) into **mechanical motion**. Controlled by an electro-hydraulic valve.

Advantages: Very high force/torque output; fast response; high power-to-weight ratio.

Disadvantages: Requires hydraulic power supply; risk of leakage; maintenance intensive.

Hydraulic Actuator Transfer Function

For a hydraulic cylinder with spool valve:

$$\frac{X(s)}{I(s)} = \frac{K_h}{s(Ms + B)}$$

where K_h = hydraulic gain, M = load mass, B = viscous damping, X = piston

displacement, I = input current to valve.

6.3 Electro-Pneumatic Valve

Electro-Pneumatic Valve: Converts an **electrical signal** into a **pneumatic pressure or flow**. Used to control pneumatic actuators. Types: I/P converter (current-to-pressure), solenoid valve.

I/P Converter: Converts 4–20 mA current signal to 3–15 psi (or 0.2–1.0 bar) pneumatic output. Standard interface between electronic controllers and pneumatic actuators in process industries.

6.4 Pneumatic Actuator

Pneumatic Actuator: Converts **compressed air pressure** into **mechanical motion** (linear or rotary). Controlled by an electro-pneumatic valve or I/P converter.

Types:

- **Spring-return diaphragm actuator:** Air pressure acts on a diaphragm against a spring. Used for control valves. Fail-safe (spring returns to safe position on air failure).
- **Double-acting cylinder:** Air applied alternately to both sides of a piston for bidirectional motion.
- **Rotary vane actuator:** Converts air pressure to rotary motion (limited angle).

Feature	Hydraulic	Pneumatic	Electric
Power density	Very high	Medium	Medium
Speed	Fast	Very fast	Fast
Precision	High	Moderate	Very high
Maintenance	High (leaks)	Low	Low
Compressibility	Incompressible	Compressible	N/A
Typical use	Heavy machinery	Process valves	Robotics, CNC

Quick Revision — Module 7

Instrumentation & Actuators | EC601

Key Formulas at a Glance

CRO voltage measurement	$V_{pk} = n_{div} \times V/div$
CRO time period	$T = n_{div} \times \text{time/div}$
CRO frequency	$f = 1/T$
CRO phase difference	$\phi = (\Delta t/T) \times 360^\circ$
Lissajous phase angle	$\sin \theta = y_1/y_2$
Lissajous frequency ratio	$f_y/f_x = N_h/N_v$
LVDT output	$V_{out} = V_{S1} - V_{S2}$; zero at null position
DC tachometer output	$V_{out} = K_t \omega$
DC servomotor TF	$\Omega(s)/V_a(s) = (K_T/R_a)/(Js + B + K_T K_b/R_a)$
AC servomotor TF	$G(s) = K_m/[s(\tau_m s + 1)]$
Hydraulic actuator TF	$X(s)/I(s) = K_h/[s(Ms + B)]$
I/P converter range	4–20 mA $\rightarrow$ 3–15 psi (0.2–1.0 bar)

One-Line Definitions

- **CRO:** Displays voltage vs time waveforms; measures amplitude, frequency, phase, rise time.
- **Trigger circuit:** Synchronises CRO sweep with input signal for stable display.
- **Delay line:** Delays signal in CRO so sweep starts before signal arrives.
- **Lissajous Figure:** Pattern displayed on CRO by two orthogonal sinusoidal signals.
- **Chop Mode:** Switching between channels at high speed; used for low-frequency signals.
- **Alternate Mode:** Switching channels after each sweep; used for high-frequency signals.
- **DSO:** Digital Storage Oscilloscope; digitises, stores, and displays waveforms.
- **Pre-trigger Viewing:** DSO feature to view signal before trigger using circular buffer.
- **Wave analyzer:** Measures amplitude of individual harmonics; narrow tunable BPF.
- **Spectrum analyzer:** Displays full frequency spectrum (amplitude vs frequency) simultaneously.

- **LVDT:** Inductive displacement sensor; output voltage proportional to core displacement.
- **Null position (LVDT):** Core centred; $V_{S1} = V_{S2}$; $V_{out} = 0$.
- **DC servomotor:** DC motor for servo use; controlled by armature voltage; linear torque-speed.
- **AC servomotor:** 2-phase induction motor; high rotor resistance; low inertia; drag-cup rotor.
- **Tachogenerator:** Speed-to-voltage transducer; used as velocity feedback sensor.
- **Electro-hydraulic valve:** Converts electrical signal to hydraulic flow; flapper-nozzle type.
- **Hydraulic servomotor:** High-force actuator; converts hydraulic power to mechanical motion.
- **I/P converter:** Converts 4–20 mA to 3–15 psi; interfaces electronic controller to pneumatic actuator.
- **Pneumatic actuator:** Converts compressed air to mechanical motion; spring-return is fail-safe.

Frequently Asked / Viva Questions

Q1. What is the function of the trigger circuit in a CRO?

It synchronises the time base sweep with the input signal so that the waveform appears stationary (stable) on the screen.

Q2. Why is a delay line used in a CRO?

To delay the signal slightly so the sweep generator starts before the signal arrives, ensuring the leading edge of the waveform is displayed.

Q3. How do you measure frequency using a CRO?

Measure the time period T (horizontal divisions $\times$ time/div for one complete cycle), then $f = 1/T$.

Q4. How can phase difference and frequency ratio be measured using Lissajous figures?

Phase difference: for equal frequencies, $\theta = \sin^{-1}(y_1/y_2)$ from the ellipse intercepts. Frequency ratio: for unequal frequencies, $f_y/f_x = N_h/N_v$, where N_h and N_v are horizontal and vertical tangency counts.

Q5. What is the difference between Chop and Alternate modes in a Dual-Trace CRO?

Chop mode switches between channels rapidly during a single sweep (best for low frequencies). Alternate mode sweeps Channel A completely, then Channel B on the next sweep (best for high frequencies to prevent trace flicker).

Q6. What is the main advantage of a DSO over an analog CRO?

Frequently Asked / Viva Questions (contd.)

DSOs digitise and store signals in digital memory, offering infinite waveform storage, pre-trigger viewing (circular buffer recording), and built-in microprocessor-based signal processing.

Q7. How does a DSO achieve pre-trigger viewing?

It continuously writes ADC samples to a circular memory buffer. When the trigger occurs, it stops writing after a specific delay, retaining the samples that were stored prior to the trigger event.

Q8. What is the difference between a wave analyzer and a spectrum analyzer?

Wave analyzer measures one frequency component at a time (narrow tunable BPF). Spectrum analyzer displays all frequency components simultaneously on screen.

Q9. What is LVDT and how does it work?

Linear Variable Differential Transformer. AC excitation on primary induces EMF in two secondary coils connected in series opposition. Output voltage is proportional to core displacement; zero at null position.

Q10. What is the output of LVDT at null position?

Zero, because the two secondary voltages are equal and opposite ($V_{S1} = V_{S2}$, so $V_{out} = V_{S1} - V_{S2} = 0$).

Q11. What are the advantages of LVDT?

Infinite resolution (frictionless/contactless), high accuracy, no mechanical wear, rugged, works in harsh environments.

Q12. What is the difference between DC and AC servomotor?

DC: controlled by armature voltage, uses brushes, high power. AC: 2-phase induction, brushless, high rotor resistance for linear torque-speed, low inertia drag-cup rotor.

Q13. What is a tachogenerator?

A speed transducer that generates a voltage proportional to rotational speed ($V = K_t \omega$). Used as velocity feedback in closed-loop speed control.

Q14. What is an electro-hydraulic servo valve?

Converts electrical current to hydraulic flow using a torque motor, flapper-nozzle, and spool valve. Controls hydraulic actuator position/speed.

Q15. What is an I/P converter?

Current-to-pressure converter. Converts 4–20 mA electrical signal to 3–15 psi pneumatic signal. Standard interface between electronic controllers and pneumatic actuators.

Q16. Why is a spring-return pneumatic actuator called fail-safe?

On loss of air supply, the spring returns the actuator (and connected valve) to a predetermined safe position (fully open or fully closed).